Terraform Tasks - 04

- Execute the tasks in the video.
- Versions:

```
PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform init
Initializing the backend...
Initializing provider plugins...
- Reusing previous version of hashicorp/random from the dependency lock file
- Reusing previous version of hashicorp/local from the dependency lock file
- Using previously-installed hashicorp/random v3.8.1
- Using previously-installed hashicorp/local v2.7.0
```

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```
terraform {
  required_providers {
    local = {
      source = "hashicorp/local"
      version = "2.6.2"
    }
  }
}

resource "random_pet" "my_pet" {
  prefix    = "Mr"
  separator = "."
  length    = var.length
}
```

-

```
PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform init -upgrade
Initializing the backend...
Initializing provider plugins...
- Finding hashicorp/local versions matching "2.6.2"...
- Finding latest version of hashicorp/random...
- Installing hashicorp/local v2.6.2...
- Installed hashicorp/local v2.6.2 (signed by HashiCorp)
- Using previously-installed hashicorp/random v3.8.1
Terraform has made some changes to the provider dependency selections recorded
in the .terraform.lock.hcl file. Review those changes and commit them to your
version control system if they represent changes you intended to make.

Terraform has been successfully initialized!
```

-

```
terraform {
  required_providers {
    local = {
      source = "hashicorp/local"
      version = "< 2.6.2"
    }
  }
}

resource "random_pet" "my_pet" {
  prefix    = "Mr"
  separator = "."
```

-

Terraform Tasks - 04

- ```
  PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform init -upgrade
  Initializing the backend...
  Initializing provider plugins...
  - Finding hashicorp/local versions matching "< 2.6.2"...
  - Finding latest version of hashicorp/random...
  - Installing hashicorp/local v2.6.1...
  - Installed hashicorp/local v2.6.1 (signed by HashiCorp)
  - Using previously-installed hashicorp/random v3.8.1
  Terraform has made some changes to the provider dependency selections recorded
  in the .terraform.lock.hcl file. Review those changes and commit them to your
  version control system if they represent changes you intended to make.
  ```
- ```
  terraform {
    required_providers {
      local = {
        source = "hashicorp/local"
        version = "!= 2.6.2"
      }
    }
  }

  resource "random_pet" "my_pet" {
    prefix    = "Mr"
    separator = "."
    length    = var.length
  ```
- ```
  PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform init -upgrade
  Initializing the backend...
  Initializing provider plugins...
  - Finding hashicorp/local versions matching "!= 2.6.2"...
  - Finding latest version of hashicorp/random...
  - Installing hashicorp/local v2.7.0...
  - Installed hashicorp/local v2.7.0 (signed by HashiCorp)
  - Using previously-installed hashicorp/random v3.8.1
  Terraform has made some changes to the provider dependency selections recorded
  in the .terraform.lock.hcl file. Review those changes and commit them to your
  version control system if they represent changes you intended to make.
  ```
- Data Sources:
- ```
  main.tf > output "hero_quote"
  resource "local_file" "multiverse" {
    filename = "multiverse.txt"
    content  = "With great power comes great responsibility."
  }

  data "local_file" "hero" {
    filename = local_file.multiverse.filename
  }

  output "hero_quote" {
    value = data.local_file.hero.content
  }
  ```

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- ```
  PS C:\Users\Viqaas\Desktop\Testing\Terraform> terraform output
  hero_quote = "With great power comes great responsibility."
  PS C:\Users\Viqaas\Desktop\Testing\Terraform>
  ```
- Meta-Arguments:
- main.tf > resource "local_file" "multiverse"

  ```
  resource "local_file" "multiverse" {
    count    = length(var.filename)
    filename = var.filename[count.index]
    content  = "I can do this all day."
  }
  ```
- variables.tf > variable "filename"

  ```
  variable "filename" {
    default = [
      "Captain America",
      "Steve Rogers",
      "Bucky Barnes"
    ]
  }
  ```
- ```
  hanges to Outputs:
   - hero_quote = "With great power comes great responsibility." -> null
  ocal_file.multiverse[0]: Destroying... [id=33b595c85d4349a6ffaeedfd24a08652c292edf3]
  ocal_file.multiverse[0]: Destruction complete after 0s
  ocal_file.multiverse[2]: Creating...
  ocal_file.multiverse[1]: Creating...
  ocal_file.multiverse[1]: Creation complete after 0s [id=edb05c6935d2179965dea811b814e5915b276297]
  ocal_file.multiverse[2]: Creation complete after 0s [id=edb05c6935d2179965dea811b814e5915b276297]
  ocal_file.multiverse[0]: Creating...
  ocal_file.multiverse[0]: Creation complete after 0s [id=edb05c6935d2179965dea811b814e5915b276297]
  ```
- main.tf > resource "local_file" "multiverse"

  ```
  resource "local_file" "multiverse" {
    for_each = var.filename

    filename = each.value
    content  = "I can do this all day."
  }
  ```
-

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- variables.tf > variable "filename"

```
variable "filename" {
  type = set(string)
  default = [
    "Captain America",
    "Steve Rogers",
    "Bucky Barnes"
  ]
}
```

-

```
Plan: 3 to add, 0 to change, 0 to destroy.
local_file.multiverse["Captain America"]: Creating...
local_file.multiverse["Bucky Barnes"]: Creating...
local_file.multiverse["Steve Rogers"]: Creating...
local_file.multiverse["Captain America"]: Creation complete after 0s [id=edb05c6935d2179965dea811b814e5915b276297]
local_file.multiverse["Steve Rogers"]: Creation complete after 0s [id=edb05c6935d2179965dea811b814e5915b276297]
local_file.multiverse["Bucky Barnes"]: Creation complete after 0s [id=edb05c6935d2179965dea811b814e5915b276297]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
```

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```
resource "aws_iam_user" "Admin-user" {
  name = "Steve"

  tags = {
    description = "Technical Team Lead"
  }
}

resource "aws_iam_policy" "adminuser" {
  name = "AdminUsers"

  policy = <<EOF
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "1234567890",
      "Effect": "Allow",
      "Action": "*",
      "Resource": "*"
    }
```

-

```
Plan: 3 to add, 0 to change, 0 to destroy.
aws_iam_user.Admin-user: Creating...
aws_iam_policy.adminuser: Creating...
aws_iam_user.Admin-user: Creation complete after 1s [id=Steve]
aws_iam_policy.adminuser: Creation complete after 2s [id=arn:aws:iam::526018540742:policy/AdminUsers]
aws_iam_user_policy_attachment.lucy-admin-access: Creating...
aws_iam_user_policy_attachment.lucy-admin-access: Creation complete after 0s [id=Steve-20260302075911950400000001]

Apply complete! Resources: 3 added, 0 changed, 0 destroyed.
PS C:\Users\Viqaas\Desktop\Testing\Terraform>
```

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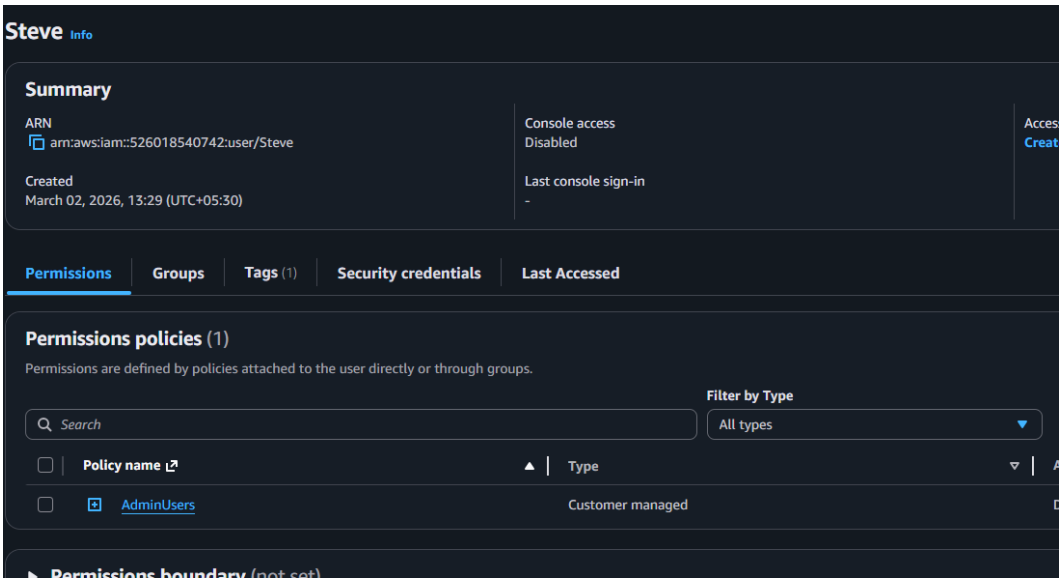
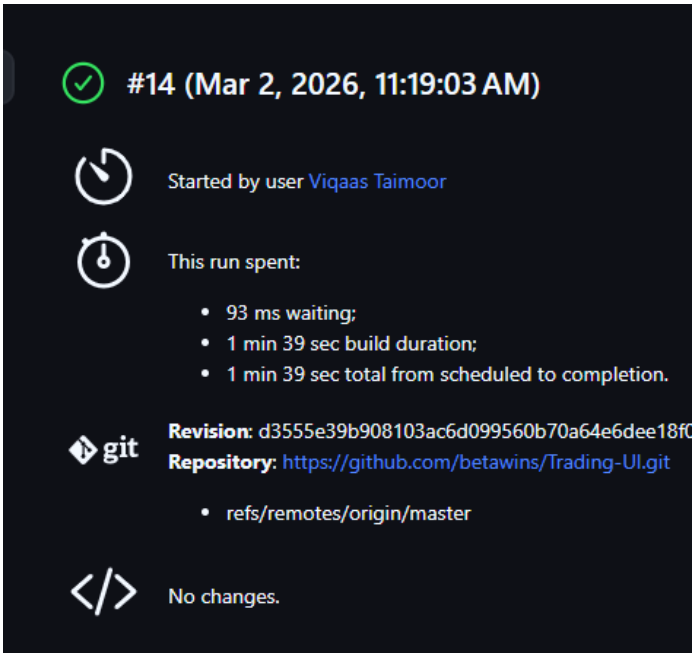
- 

- Integrate **Terraform in Jenkins** using the **Terraform plugin**.
- 

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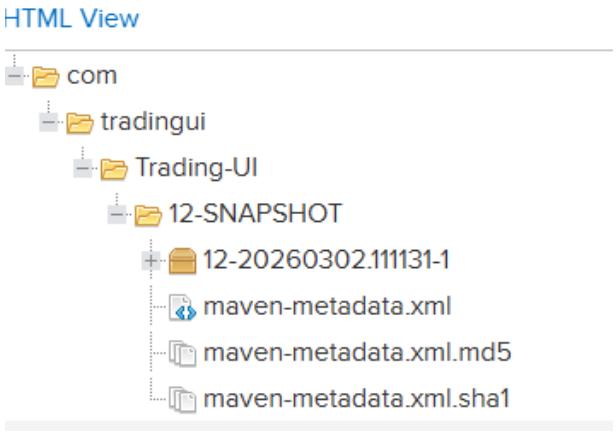
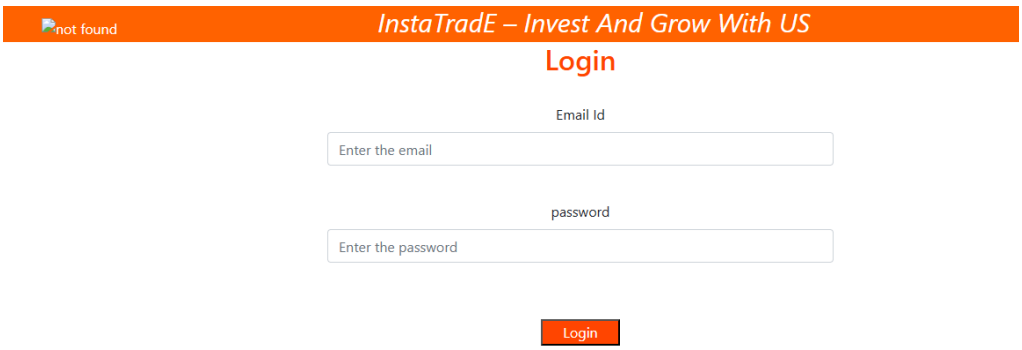
- 



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- Explain **10 Maven commands**.
    - mvn clean - removes previous build artifacts.
    - mvn compile - compiles source code.
    - mvn test - runs unit tests without packaging.
    - mvn package - creates JAR or WAR file.
    - mvn install - installs artifact into local Maven repo.
    - mvn deploy - uploads artifact to remote repository (like Nexus).
    - mvn clean install – it cleans > compiles > tests > packages > installs.
    - mvn validate - checks if project is ready to build.
    - mvn clean package - deletes old build and creates new JAR/WAR.
    - mvn clean deploy - builds and publishes the artifact to a remote repository.